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The Editors
Computer Methods in Applied Mechanics and Engineering

Dear Editors,

I submit the manuscript *"Operator-adapted learned multigrid cycles for high-contrast elliptic problems: accuracy, failure modes and cost"* for consideration as a research article in *Computer Methods in Applied Mechanics and Engineering*.

The paper is a controlled methodological study. It asks what learned smoothing adds to one pass of a multigrid cycle for high-contrast diffusion problems on unstructured finite element meshes, when the transfers, Galerkin coarse operators and near-kernel deflation vectors are computed from each sample's operator and only the smoothers are learned. The architecture provably represents the classical cycles with fixed polynomial smoothers on the same hierarchy, is positively homogeneous in the load, and has an energy-norm safeguard. This allows a paired comparison with a classical cycle that receives exactly the same hierarchy, coarse solution, deflation and output projection.

At four update stages per smoother, the learned pass is about ten times more accurate in the energy norm than the degree-four Chebyshev-smoothed pass on conductive channels and about twice as accurate on insulating barriers, reproducibly across held-out data seeds and training runs; ablations attribute parts of this gain to the operator-adapted transfers, deflation, Galerkin output and nonlinearity. The paper reports with equal weight where the result stops: the margin shrinks quickly when the classical pass receives more smoothing, and three classical passes are more accurate at a third of the cost; a network trained with one load loses its advantage on other loads; floating islands keep substantial error; and the implementation is 12–98 times slower than converged classical solvers at matched accuracy. The manuscript therefore does not claim a fast solver or a broadly superior neural operator. Its contribution is a reproducible measurement of what learned smoothing contributes inside a prescribed multilevel pass, and of why that contribution does not yet translate into solver efficiency, which I believe is useful to readers designing learned components for multilevel solvers in computational mechanics.

All tables are regenerated from stored per-sample records by scripts that, with the code, records and checkpoints, are available from me on request. The manuscript has not been published and is not under consideration elsewhere. The use of generative AI tools in preparing this work is declared

in the manuscript. There are no competing interests and no specific funding.

Yours sincerely,

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